HURRICANE ELECTRIC
INTERNET SERVICES

# Peering Security

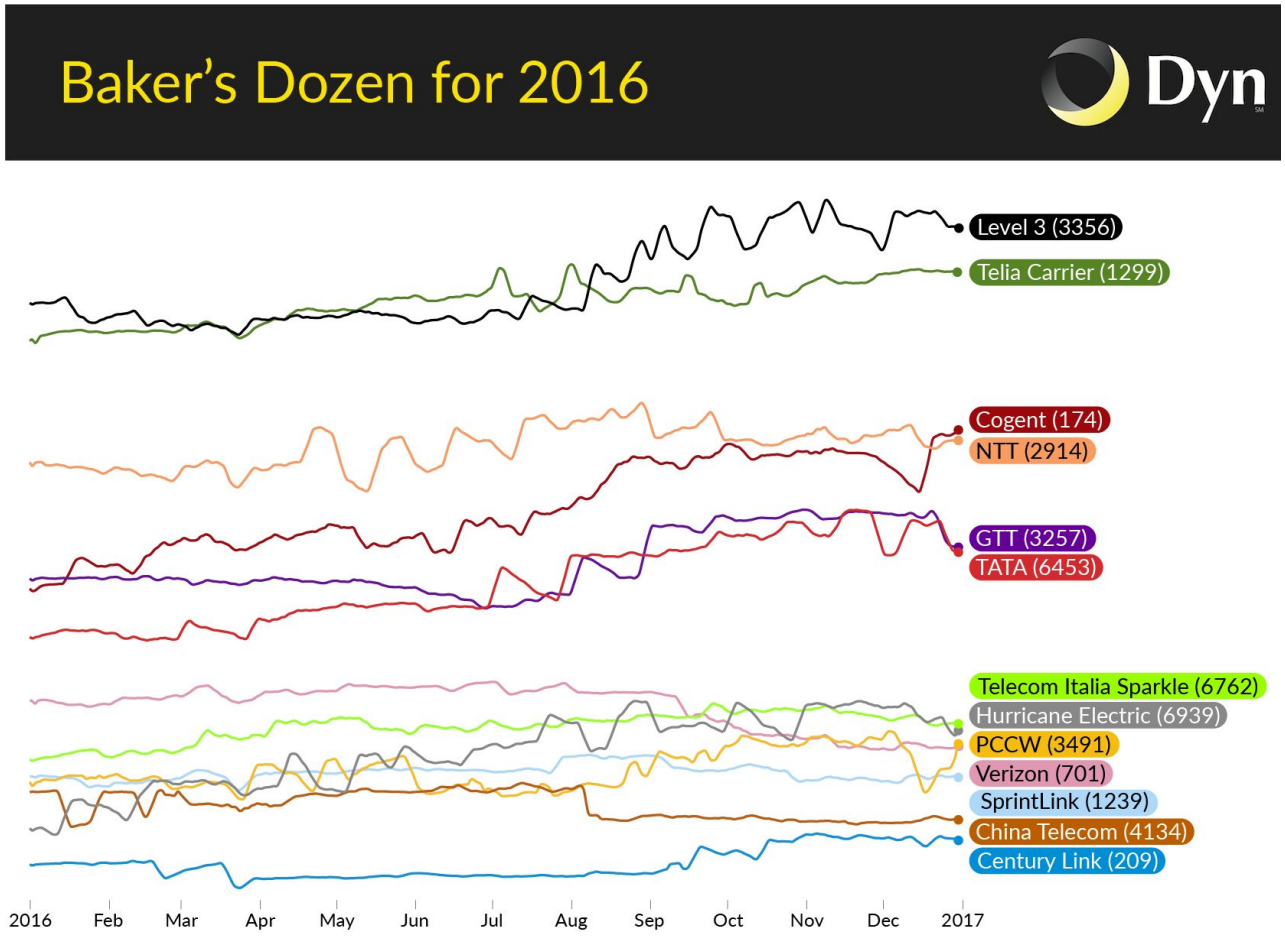
NZNOG
2019

Walt Wollny, Director Interconnection Strategy
Hurricane Electric  AS6939

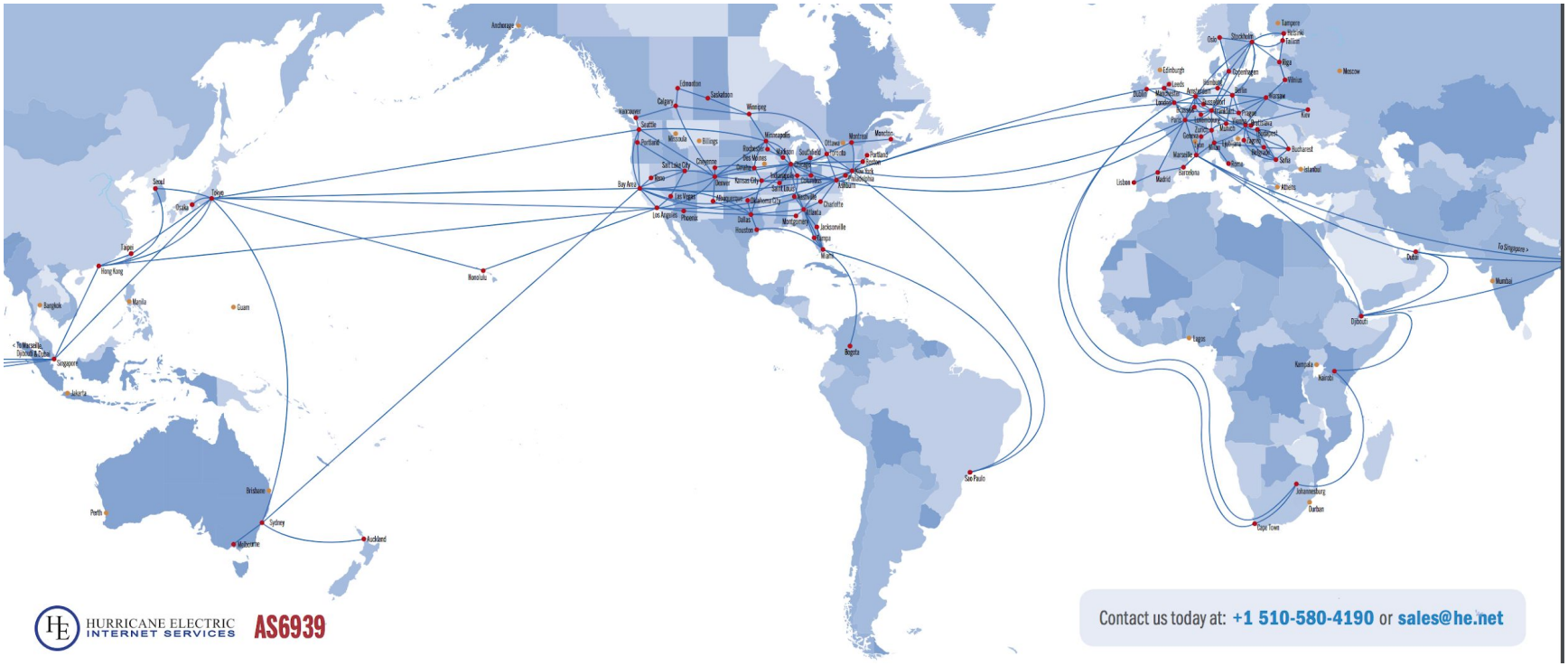
# Who is Walt Wollny?

- Hurricane Electric AS6939 – 4 years
  - Director Interconnection Strategy – supporting the network to reach to over 42 counties and over 206 Internet Exchanges and 184 POPS. Focus on Global connectivity.
- Amazon AS16509 – 4 years
  - Developed IP Transit and Peering on five continents.
  - Primary focus on Japan, Singapore, Hong Kong, India, Taiwan, Philippines, Australia.
  - Over 62 new CDN sites.
- Microsoft AS8075 – 13 years
  - Developed IP Transit and Peering on four continents.
  - Primary focus on US, EU and South America.

# Hurricane Electric #8 in 2017 up from #13 in 2016 in the Global IPv4 rankings!



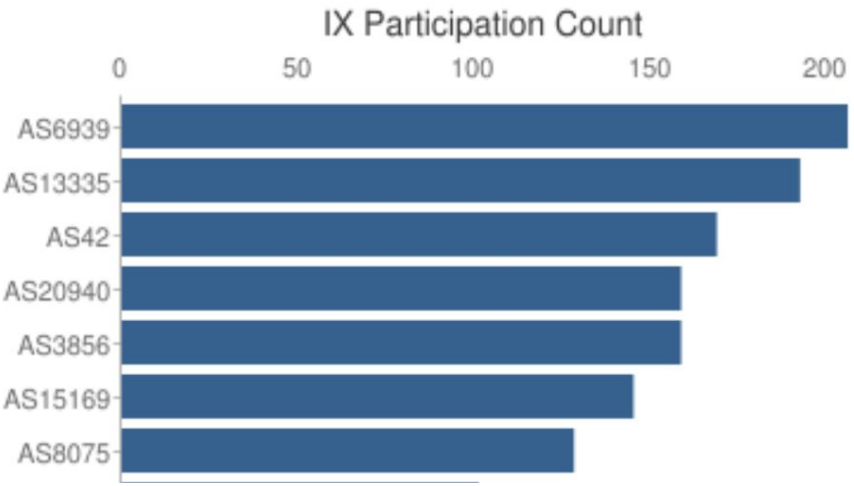
# Hurricane Electric Backbone



# The Most Peering Exchanges

Internet Exchanges | Exchange Participants

| IX Participation Count | | |
|---|---|---|
| **ASN** | **Name** | **IXes** |
| AS6939 | Hurricane Electric LLC | 206 |
| AS13335 | Cloudflare, Inc. | 195 |
| AS42 | WoodyNet | 169 |
| AS20940 | Akamai International B.V. | 161 |
| AS3856 | Packet Clearing House | 160 |
| AS15169 | Google LLC | 148 |
| AS8075 | Microsoft Corporation | 130 |



# What does security have to do with Peering?

A lot. Now.

Security was an afterthought, but it can’t be a forgotten thought.

Some of the basics...

# Basics

- Best defenses for your network?
  - Logical Port Security
  - Routing Security
  - Best Practices

# Basics - Port Security

- Many IXPs will post their recommended port configuration (HKIX, AMS-IX, etc )
- Don’t just connect a default interface to an IX Port!
- Services like Proxy-ARP will both disrupt the IX, as well as degrade your own network

# Basics - Port Security

- Don’t forget to apply ACL’s to your interfaces!
- The SIX (Seattle Internet Exchange) has a great example here.
- Your IXP port is an exposed piece of your network
- Hundreds of other networks are directly connected
- Remove this security risk!

# Basics - Port Security

- Your IXP Port is a target for DDoS Attacks!
- Applying the best security practices will help keep your network online during attacks.
- Part of the security is in making sure you know what your IX Provider is doing too...

# Basics - Routing Security

- The IXP is responsible for protecting the infrastructure
- The IX LAN is not your IP space and should not be routed.
- Checking this...

# Basics - Routing Security

Public Peering Exchange Points

JPNAP

| Exchange<br>ASN | IPv4<br>IPv6 | Speed<br>RS Peer |
|---|---|---|
| JPNAP Osaka<br>6939 | 210.173.178.70<br>2001:7fa:7:2::6939:1 | 10G<br>○ |
| JPNAP Tokyo<br>6939 | 210.173.176.106<br>2001:7fa:7:1::6939:1 | 10G<br>○ |

# Basics - Routing Security

https://bgp.he.net/ip/210.173.176.106

Apps | TPE | golf | HE stuff | personal | toons | ITW | Matrix - ITA Softw... | Google C

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Search

**210.173.176.106**

**Quick Links**

- BGP Toolkit Home
- BGP Prefix Report
- BGP Peer Report
- Exchange Report
- Bogon Routes
- World Report
- Multi Origin Routes
- DNS Report
- Top Host Report
- Internet Statistics
- Looking Glass
- Network Tools App
- Free IPv6 Tunnel

IP Info | Whois | DNS | RBL

210.173.176.106 (gigabitethernet2-8.core1.tyo1.he.net)

| Announced By | | |
|---|---|---|
| **Origin AS** | **Announcement** | **Description** |
| AS7521 | 210.173.160.0/19 | |
| AS7521 | 210.173.176.0/20 | |
| AS18403 | 210.173.176.0/24 | |

Address has 0 hosts associated with it.

# Basics - Routing Security

Oceania

| CC | Exchange | Speed | IPv4 | IPv6 |
|---|---|---|---|---|
| AU | Equinix Melbourne | 10GE | 183.177.61.28 | 2001:de8:6:1::6939:1 |
| AU | Equinix Sydney | 10GE | 45.127.173.24 | 2001:de8:6::6939:1 |
| AU | NSW-IX Sydney | 10GE | 218.100.52.249 | 2001:7fa:11:4:0:1b1b:0:1 |
| AU | VIC-IX Melbourne | 10GE | 218.100.78.108 | 2001:7fa:11:1:0:1b1b:0:1 |
| AU | MegaIX Melbourne | 10GE | 103.26.71.122 | 2001:dea:0:30::7a |
| AU | MegaIX Sydney | 10GE | 103.26.68.236 | 2001:dea:0:10::ec |
| NZ | APE | 10GE | 192.203.154.197 | 2001:7fa:4:c0cb::9ac5 |
| NZ | AKL-IX | 10GE | 43.243.21.17 | 2001:7fa:11:6:0:1b1b:0:1 |
| NZ | MegaIX Auckland | 10GE | 43.243.22.82 | 2001:dea:0:40::52 |

# Basics - Routing Security

- Make sure the IX LAN is not routed on the internet.
- bgp.he.net is a quick and easy way to check
- Please don’t originate the IXP LANs!

- There are many IXPs LANS that are being routed today
- Please help eliminate this risk

# Basics - Routing Security

- https://bgpmon.net/
  - Monitoring, notifications of when errors occurs
  - Price is free for up to five prefixes per month.

# BGPmon.net Notification

BGPmon Alert

**Sent:** Wednesday, January 30, 2019 at 11:08 AM

**To:** info@seattleix.net

You received this email because you are subscribed to BGPmon.net.
For more details about these updates please visit:
https://portal.bgpmon.net/myalerts.php

==========
Possible Prefix Hijack (Code: 10)
==========

| | |
|---|---|
| Your prefix: | 206.81.80.0/22: |
| Update time: | 2019-01-29 21:55 (UTC) |
| Detected by #peers: | 1 |
| Detected prefix: | 206.81.80.0/23 |
| Announced by: | AS10310 (YAHOO-1 - Yahoo!, US) |
| Upstream AS: | AS29467 (LUXNETWORK Network Service Provider in Luxembourg, LU) |
| ASpath: | 60983 29467 10310 |
| Alert details: | https://portal.bgpmon.net/alerts.php?details&alert_id=86973730 |
| Mark as false alert: | https://portal.bgpmon.net/fp.php?aid=86973730 |

---
*for questions regarding the change code or other question, please see:
https://portal.bgpmon.net/faq.php

Latest BGPmon news: http://bgpmon.net/blog/
* Popular Destinations rerouted to Russia
* Todayâ€™s BGP leak in Brazil
* BGP leak causing Internet outages in Japan and beyond.

# Basics - Routing Security

- Why we should care
  - The DDoS That Almost Broke the Internet
  - Cloudflare March 2013 ~120Gbps attack on LINX

# Basics - Routing Security

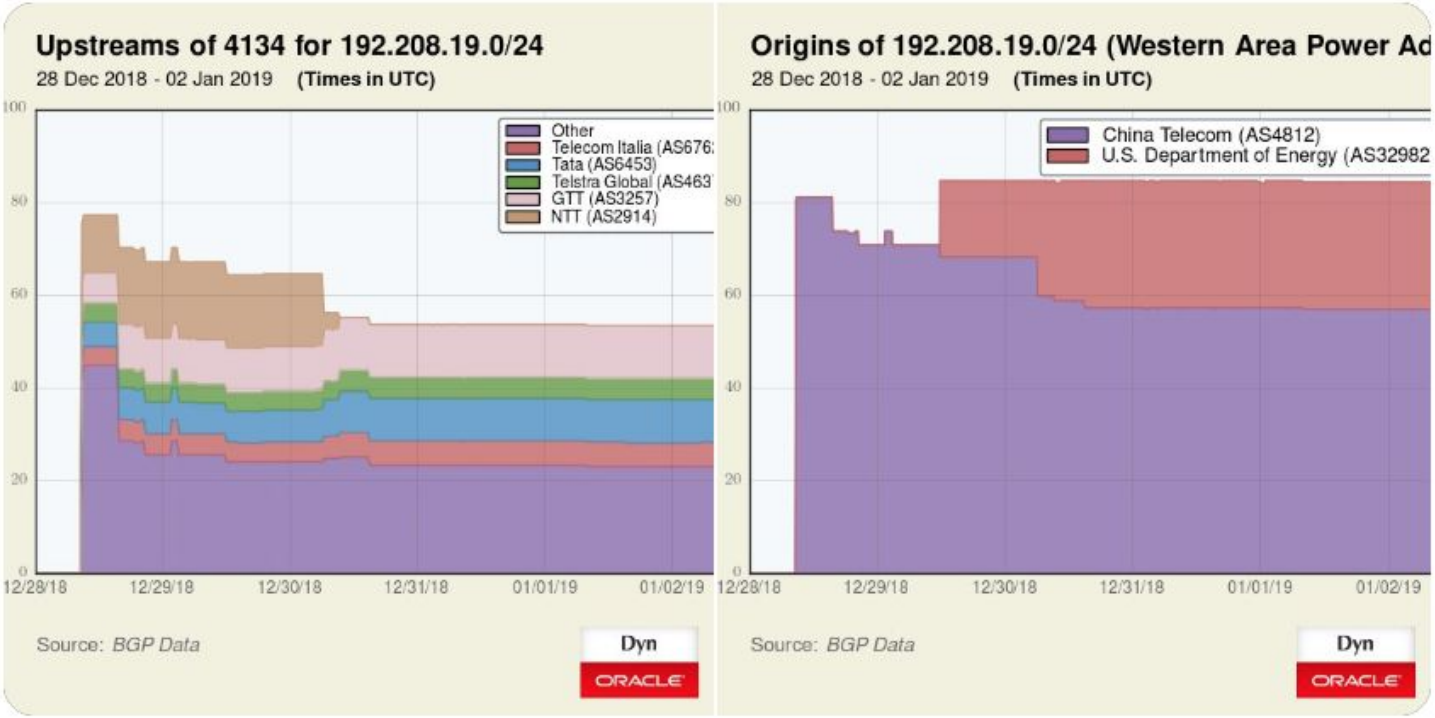
You <u>must</u> filter your peers.

**InternetIntelligence**
@InternetIntel

Following

Replying to @InternetIntel @bad_packets

China Telecom hijack of US Dept of Energy route continuing into 6th day. Both entities are presently announcing the same Dept of Energy prefix (192.208.19.0/24).

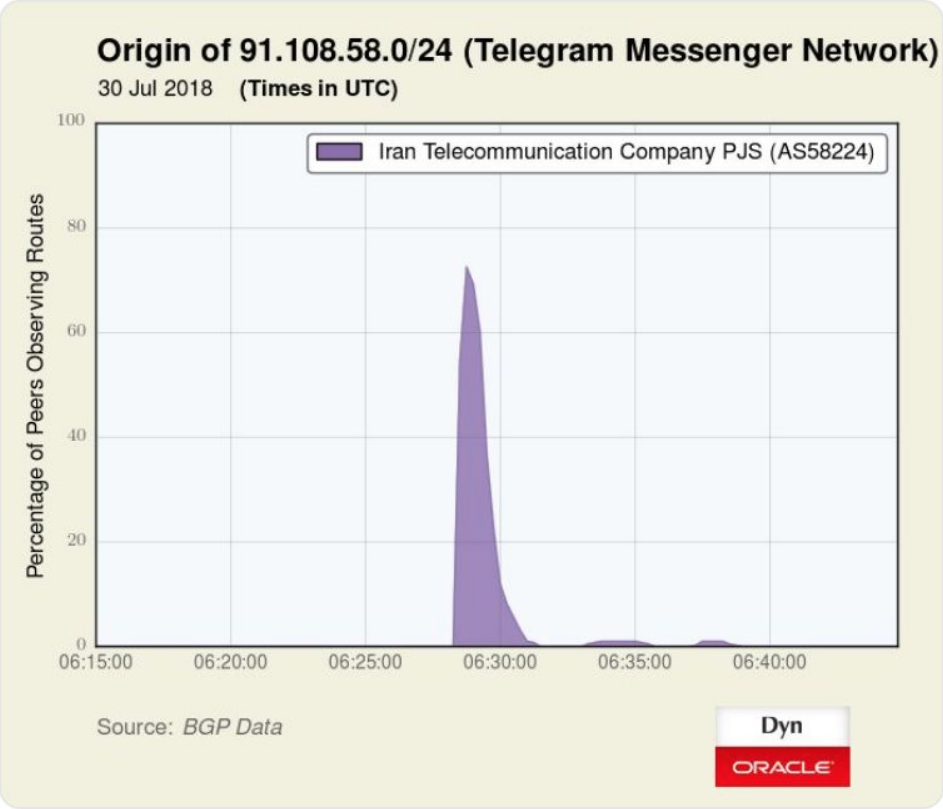



6:11 AM - 2 Jan 2019

**InternetIntelligence**
@InternetIntel

Follow

At 06:28 UTC earlier today (30-Jul), an Iranian state telecom network briefly leaked over 100 prefixes. Most were Iranian networks, but the leak also included 10 prefixes of popular messaging app @telegram (8 were more-specifics).




7:45 AM - 30 Jul 2018

# Basics - Routing Security

- Routing security is important in two directions:
  - The routes you receive
  - The routes you announce

- Starting with the routes you receive…

# Basics - Routing Security

- Most networks won't/don't filter their peers.
- This is negligent behavior
- The routes you receive should be filtered in three ways:
  - Prefix Count
  - AS-Path
  - Prefix list

# http://routing.he.net

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OUTE FILTERING HOME ALGORITHM

# AS13335

| ASN | STATUS | PEERINGDB_IRR | EXTRACTED_V4 | EXTRACTED_V6 | OK_V4 | OK_V6 | SOURCE |
|---|---|---|---|---|---|---|---|
| 13335 | explicit | AS-CLOUDFLARE | | | AS-CLOUDFLARE | AS-CLOUDFLARE | peeringdb |

## ILTERS

| AF | AS-SET NAME | IRR STATUS | IRR BUILT | IRR LINES | PREFIXES RECEIVED | FILTER BUILT | FILTER LINES | POLICY | REASONS | FILTE |
|---|---|---|---|---|---|---|---|---|---|---|
| 4 | AS-CLOUDFLARE | good | October 18 2018 13:18:53 | 1203 | 522 | October 19 2018 13:18:44 | 522 | DISPLAY | DISPLAY | DISPLA |
| 6 | AS-CLOUDFLARE | good | October 18 2018 13:19:08 | 553 | 108 | October 19 2018 13:18:47 | 108 | DISPLAY | DISPLAY | DISPLA |

## REFIX LISTS

| AF | ROUTER | NAME | STATUS | CHECKED | EXISTING_LINES | VERIFIED | EXISTING | DELTA | LOG |
|---|---|---|---|---|---|---|---|---|---|
| 4 | core1.ams1.he.net | prefix-filter-as13335 | verified | July 02 2018 15:23:00 | 522 | July 02 2018 15:23:01 | DISPLAY | DISPLAY | DISPLA |

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ROUTE FILTERING HOME ALGORITHM

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# http://routing.he.net

## SESSIONS

295 sessions.

SESSION STATUS IS NON REALTIME, DATA IN TABLE IS DELAYED APPROXIMATELY 24 HOURS

| IP | ROUTER | STATUS | ACCEPTED | FILTERED | RECEIVED | RCVD STATUS | RCVD UPDATED | RCVD ACCEPTED | RCVD FILTERED |
|---|---|---|---|---|---|---|---|---|---|
| 103.16.102.93 | core1.sin1.he.net | ESTAB | 0 | 266 | DISPLAY | good | October 20 2018 01:52:05 | 0 | 266 |
| 103.231.152.33 | core1.sin1.he.net | ESTAB | 270 | 0 | DISPLAY | good | October 18 2018 18:39:16 | 270 | 0 |
| 103.246.232.134 | core1.osa1.he.net | ESTAB | 255 | 0 | DISPLAY | good | September 17 2018 00:07:52 | 255 | 0 |

# http://routing.he.net

## SESSIONS

295 sessions.

SESSION STATUS IS NON REALTIME, DATA IN TABLE IS DELAYED APPROXIMATELY 24 HOURS

| IP | ROUTER | STATUS | ACCEPTED | FILTERED | RECEIVED | RCVD STATUS | RCVD UPDATED | RCVD ACCEPTED | RCVD FILTERED |
|---|---|---|---|---|---|---|---|---|---|
| 103.16.102.93 | core1.sin1.he.net | ESTAB | 0 | 266 | DISPLAY | good | October 20 2018 01:52:05 | 0 | 266 |
| 103.231.152.33 | core1.sin1.he.net | ESTAB | 270 | 0 | DISPLAY | good | October 18 2018 18:39:16 | 270 | 0 |
| 103.246.232.134 | core1.osa1.he.net | ESTAB | 255 | 0 | DISPLAY | good | September 17 2018 00:07:52 | 255 | 0 |

```
SSH@core1.ams1.he.net>terminal length 0
sh ip bgp nei 185.1.32.22 received-routes
       There are 262 received routes from neighbor 185.1.32.22
Searching for matching routes, use ^C to quit...
Status A:AGGREGATE B:BEST b:NOT-INSTALLED-BEST C:CONFED_EBGP D:DAMPED
       E:EBGP H:HISTORY I:IBGP L:LOCAL M:MULTIPATH m:NOT-INSTALLED-MULTIPATH
       S:SUPPRESSED F:FILTERED s:STALE x:BEST-EXTERNAL
       Prefix              Next Hop        MED        LocPrf     Weight Status
1      1.0.0.0/24          185.1.32.22                100        0      ME
         AS_PATH: 13335
2      1.1.1.0/24          185.1.32.22                100        0      ME
         AS_PATH: 13335
3      23.227.63.0/24      185.1.32.22                100        0      ME
         AS_PATH: 13335
4      64.68.192.0/24      185.1.32.22                100        0      ME
         AS_PATH: 13335
5      66.235.200.0/24     185.1.32.22                100        0      EF
         AS_PATH: 13335
6      104.16.0.0/12       185.1.32.22                100        0      ME
         AS_PATH: 13335
7      104.16.0.0/20       185.1.32.22                100        0      ME
```

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SSH@core1.ams1.he.net>terminal length 0
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         AS_PATH: 13335
2      1.1.1.0/24          185.1.32.22                100        0      ME
         AS_PATH: 13335
3      23.227.63.0/24      185.1.32.22                100        0      ME
         AS_PATH: 13335
4      64.68.192.0/24      185.1.32.22                100        0      ME
         AS_PATH: 13335
5      66.235.200.0/24     185.1.32.22                100        0      EF
         AS_PATH: 13335
6      104.16.0.0/12       185.1.32.22                100        0      ME
         AS_PATH: 13335
7      104.16.0.0/20       185.1.32.22                100        0      ME
```

```
Toms-MacBook-Pro-38:Downloads tom$ whois -h whois.radb.net 66.235.200.0
route:      66.235.200.0/24
descr:      CMI  (Customer Route)
origin:     AS38082
mnt-by:     MAINT-AS58453
changed:    qas_support@cmi.chinamobile.com 20180906
source:     RADB

route:      66.235.200.0/24
descr:      CMI IP Transit
origin:     AS38082
admin-c:    MAINT-CMI-INT-HK
tech-c:     MAINT-CMI-INT-HK
mnt-by:     MAINT-CMI-INT-HK
changed:    qas_support@cmi.chinamobile.com 20180906
source:     NTTCOM
```

# Hurricane Electric Route Filtering Algorithm

- Read more here
- http://routing.he.net/algorithm.html

- Example:
- xx.7.224.0/24,rejected,does not strictly match IRR policy or RIR handles
- xx.10.254.0/23,accepted,strictly matched IRR policy
- xx.17.248.0/24,accepted,strictly matched IRR policy
- xx.26.36.0/22,rejected,does not strictly match IRR policy or RIR handles
- xx.26.39.0/24,rejected,does not strictly match IRR policy or RIR handles

# Hurricane Electric
# Route Filtering

- Please check and update your IRR or RIR handles

- Check your routing here:
- http://routing.he.net/

- We at now filtering ~90% of all our peers
- Rolling it out slowly over the last six months

# Resources

- Seattle SIX
- Tom Paseka tom@cloudflare.com
- BGP.HE.NET
- https://bgpmon.net/
- https://twitter.com/InternetIntel
- DYN

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# Thanks!

Walt Wollny, Director Interconnection Strategy
Hurricane Electric  AS6939
walt@he.net